Republic of the Philippines
PARTIDO STATE UNIVERSITY
Camarines Sur

"Community Disaster Information and Emergency Contact System"

MEMBERS

Alerta, Ayesa P.

Pabon, Jovert A.

Paladan, Kirby H.

Zamudio, Laica Jene B.

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I. Project Background

Project Overview

The Philippines is consistently ranked as one of the world's most disaster-prone countries, positioned along the Pacific Ring of Fire and the typhoon belt. The nation faces20+ annual typhoons, frequent earthquakes, volcanic eruptions, and flooding, causing massive economic and human losses. Key risks include landslides, tsunamis, and climate-related events. According to the National Disaster Risk Reduction and Management Council, these disasters frequently affect communities and require effective communication and coordination between local authorities and residents. However, many communities still rely on manual reporting and scattered information systems, which can delay emergency response and access to important contact information.

A Community Disaster Information and Emergency Contact System is designed to help local communities manage disaster-related information through a centralized digital database. The system stores important data such as emergency contacts, evacuation centers, community resources, and disaster alerts. By providing easy access to critical information, the system can help improve coordination, communication, and response time during emergencies.

Information systems play a vital role in disaster preparedness because they allow organizations to collect, store, and share critical information quickly among responders and the public.

Problem Statement

Despite the importance of disaster preparedness, many communities face several challenges in managing disaster information and emergency communication. Here are some of the major issues:

- Lack of a centralized system where emergency contacts and disaster-related data can be stored and accessed easily.
- Information is often scattered across different offices or maintained manually, which can lead to delays in retrieving important data during emergencies.
- The slow dissemination of information. Without an organized system, local officials may find it difficult to notify residents about evacuation orders, disaster warnings, or emergency updates.
- Manual record keeping increases the risk of outdated information, data loss, and human errors.

These challenges can reduce the efficiency of disaster response efforts and may lead to confusion among community members during critical situations.

Proposed Solution

To address these issues, this project proposes the development of a Community Disaster Information and Emergency Contact System that uses a centralized database to manage disaster-related data. The system stores important information such as emergency contacts, evacuation centers, organizations, community profiles, disaster records, and disaster announcements in a structured database.

Through database technology, the system allows authorized users (Administrators and Officers) to store, update, and retrieve information efficiently, while Residents are limited to viewing data. This improves data accuracy, reduces redundancy, and ensures data integrity.

The system also automates processes such as searching for emergency contacts, managing disaster records and announcements, and organizing evacuation center information. By implementing this system, local authorities and community members can access reliable and updated information quickly during emergency situations.

Project Objectives

- To develop a centralized database system that stores community disaster information and emergency contact details within the project implementation period.
- To improve access to emergency contact information for community members and local authorities through a digital platform.
- To reduce the time required to retrieve disaster-related information by implementing a searchable and organized database.
- To enhance communication during emergencies by providing a system that stores updated disaster alerts and emergency contacts.
- To ensure data accuracy and integrity by implementing proper database management and structured data storage.

II. Scope and Limitations

Scope

The Community Disaster Information and Emergency Contact System focuses on developing a database system that stores and manages disaster-related information for the community. The system will maintain records of emergency contacts, evacuation centers, disaster announcements, and basic community information.

The system will allow authorized users such as local disaster management officers or administrators to input, update, and manage disaster information. Community members may also access important emergency contacts and disaster-related information when needed. The system aims to improve the organization, storage, and retrieval of data to support faster disaster response and communication within the community.

The project primarily focuses on the design and implementation of a database system that ensures data accuracy, organized storage, and efficient retrieval of disaster-related information.

Limitations

This project has several limitations. First, the system will only manage disaster information and emergency contacts within a specific community or organization. It will not cover national disaster monitoring systems.

Second, the system will depend on the availability of internet or network access for users to retrieve information. During severe disasters where communication infrastructure is damaged, access to the system may be limited.

Lastly, the system will focus mainly on data management and information retrieval. It will not include advanced disaster prediction technologies or integration with external government disaster monitoring systems.

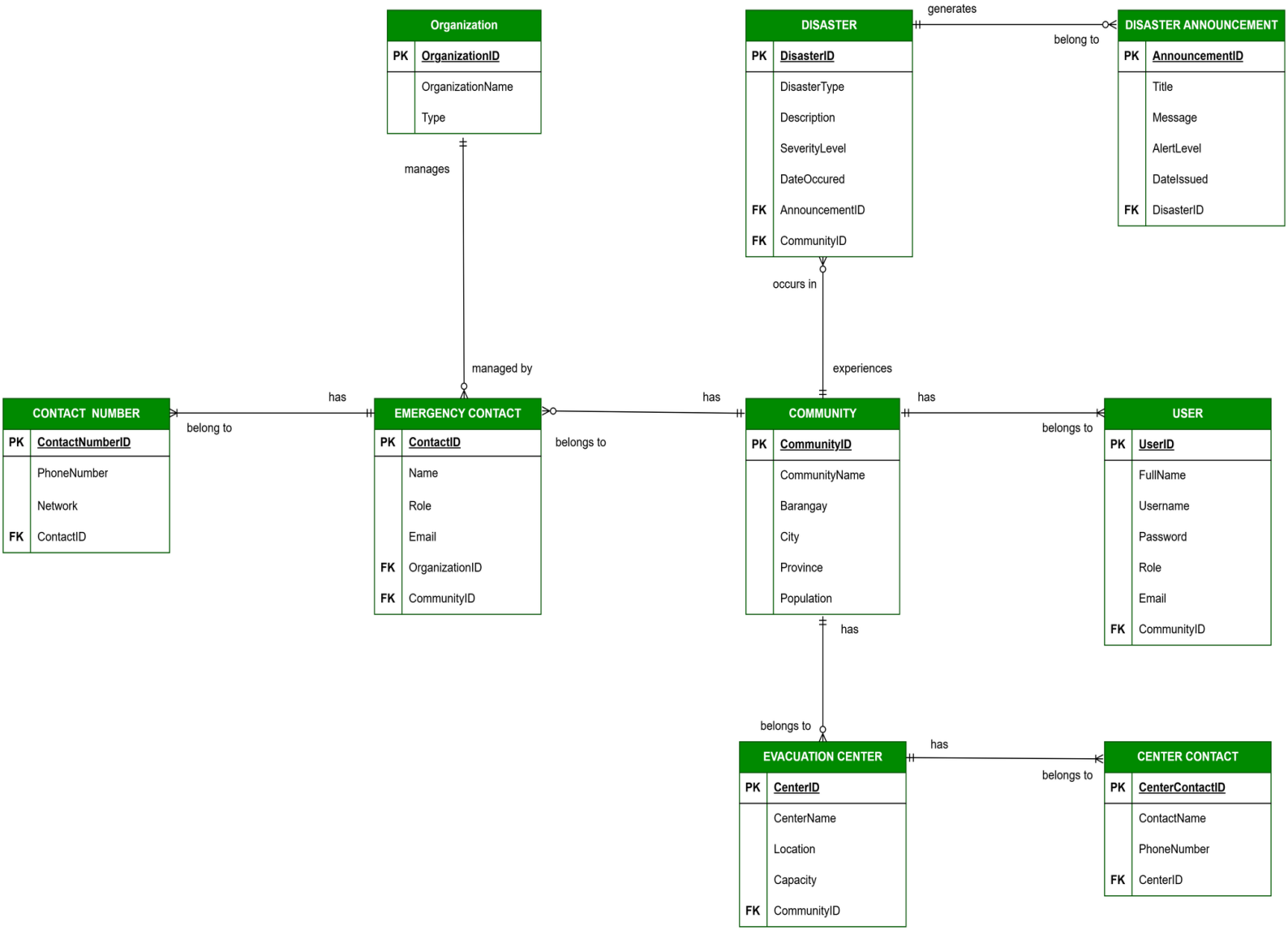
III. Business Rules

- Each COMMUNITY is uniquely identified by CommunityID.
- Each COMMUNITY must store CommunityName, Barangay, City, Province, and Population.
- Each COMMUNITY must have at least one USER
- A COMMUNITY may have zero or many, EMERGENCY CONTACTS, DISASTERS, and EVACUATION CENTERS.
- CommunityID (FK) uniquely identifies all other attributes in COMMUNITY.
- CommunityName, Barangay, City, Province collectively describe the geographic identity of the COMMUNITY.
- Population represents the number of people within the COMMUNITY.
- Each USER is uniquely identified by UserID.
- Each USER must have FullName, Username, Password, Role, and Email.
- Each USER must belong to exactly one COMMUNITY.
- A USER's Role must be one of the following: Admin, Officer, or Resident.
- UserID determines all USER attributes.
- Username and Email must be unique per USER.
- Role determines the level of access of the USER.
- CommunityID (FK) links USER to COMMUNITY.
- Each ORGANIZATION is uniquely identified by OrganizationID.
- Each ORGANIZATION must have OrganizationName and Type.
- OrganizationID uniquely identifies each ORGANIZATION.
- OrganizationName and Type describe the ORGANIZATION.
- Each EMERGENCY CONTACT is uniquely identified by ContactID.
- Each EMERGENCY CONTACT must have Name, Role, and Email.
- Each EMERGENCY CONTACT must belong to exactly one ORGANIZATION and one COMMUNITY.

- An EMERGENCY CONTACT must have at least one CONTACT NUMBER
- ContactID determines all EMERGENCY CONTACT attributes.
- Name, Role, and Email describe the identity and function of the CONTACT.
- OrganizationID (FK) links the contact to an ORGANIZATION.
- CommunityID (FK) links the contact to a COMMUNITY.
- Each CONTACT NUMBER is uniquely identified by ContactNumberID.
- Each CONTACT NUMBER must store a valid PhoneNumber and its corresponding Network.
- Each CONTACT NUMBER must belong to exactly one EMERGENCY CONTACT.
- An EMERGENCY CONTACT may have one or many CONTACT NUMBERS.
- ContactNumberID uniquely identifies the phone record.
- PhoneNumber stores the contact's reachable number.
- ContactID (FK) links the number to an EMERGENCY CONTACT.
- Each DISASTER is uniquely identified by DisasterID.
- Each DISASTER must include DisasterType, Description, SeverityLevel, and DateOccurred.
- Each DISASTER must occur in exactly one COMMUNITY.
- A COMMUNITY may experience zero or many DISASTERS.
- DisasterID identifies each DISASTER record.
- DisasterType and SeverityLevel classify the disaster.
- DateOccurred records when the disaster happened.
- CommunityID (FK) links DISASTER to COMMUNITY.
- Each DISASTER ANNOUNCEMENT is uniquely identified by AnnouncementID.
- Each announcement must include Title, Message, AlertLevel, and DateIssued.
- Each DISASTER ANNOUNCEMENT must be issued for exactly one DISASTER.
- A DISASTER may have zero or many ANNOUNCEMENTS.
- AnnouncementID identifies each announcement.

- Title and Message describe the announcement content.
- AlertLevel indicates urgency.
- DateIssued records when it was released.
- DisasterID (FK) links to the DISASTER.
- Each EVACUATION CENTER is uniquely identified by CenterID.
- Each EVACUATION CENTER must have CenterName, Location, and Capacity.
- Each EVACUATION CENTER must belong to exactly one COMMUNITY.
- A COMMUNITY may have zero or many EVACUATION CENTERS.
- CenterID uniquely identifies the center.
- CenterName and Location describe the center.
- Capacity represents maximum number of occupants.
- CommunityID (FK) links to COMMUNITY.
- Each CENTER CONTACT is uniquely identified by CenterContactID.
- Each CENTER CONTACT must include ContactName and PhoneNumber.
- Each CENTER CONTACT must belong to exactly one EVACUATION CENTER.
- An EVACUATION CENTER may have one or many CENTER CONTACTS.
- CenterContactID identifies the contact.
- ContactName stores the person in charge.
- PhoneNumber provides contact access.
- CenterID (FK) links to EVACUATION CENTER.

IV. ER-Diagram



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